A new lower bound for the mixed covering code $K(2, 5, 1)$

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Abstract

Let $K(t, b, 1)$ be the minimum size of a code in $\mathbb{F}_3^t \times \mathbb{F}_2^b$ with covering radius 1. The tables of mixed binary/ternary covering codes give $32 \le K(2, 5, 1) \le 36$, the smallest open entry for covering radius 1. We prove $K(2, 5, 1) \ge 34$. The space is split into nine blocks by the two ternary coordinates; a counting inequality leaves few possible distributions of codewords over the blocks, and each remaining case is shown infeasible by integer programming. Two independent solvers agree on every case, and every step of the reduction is checked by separately written code.

1 Introduction

A code $C \subseteq \mathbb{F}_3^t \times \mathbb{F}_2^b$ has *covering radius* 1 if every word is within Hamming distance 1 of some codeword, and $K(t, b, 1)$ is the minimum size of such a code [2]. Bounds for $t+b \le 13$ are collected in the tables of Kéri [4], following [6, 1]. For $K(2, 5, 1)$, the entry with the smallest space ($9 \cdot 32 = 288$ words) among the open radius-1 entries, the tables give

$$32 \le K(2, 5, 1) \le 36.$$

The upper bound is due to Hämäläinen and Rankinen [3], the lower bound to Lang, Quistorff and Schneider [5]. We are not aware of any later improvement.

Theorem 1. *$K(2, 5, 1) \ge 34$.*

Since a code with fewer than M words can be enlarged to exactly M words without losing the covering property, it suffices to show that no covering code with exactly M words exists for $M = 32$ and $M = 33$.

2 Method

Write words as (a, b, r) with $a, b \in \mathbb{F}_3$ and $r \in \mathbb{F}_2^5$. The *block* $c = (a, b)$ is the set of 32 words with these ternary coordinates; the nine blocks form a 3×3 grid. For a code C let $M_c = |C \cap c|$.

Lemma 2. *If C has covering radius 1, then for every block c,*

$$6M_c + \sum_{d \in N(c)} M_d \;\ge\; 32,$$

where $N(c)$ is the set of the four blocks in the same row or column of the grid as c.

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Proof. Every word of c lies in the ball of some codeword, so the balls of radius 1 around the codewords, intersected with c, cover all 32 words of c. A codeword in c meets c in itself and its 5 binary neighbours. A codeword in a block of $N(c)$ differs from c in exactly one ternary coordinate, so its ball meets c in exactly one word. A codeword in any other block differs from every word of c in both ternary coordinates, so its ball misses c. Summing gives the inequality. □

The search for a code with M words therefore proceeds in two steps.

1. **Distributions.** List every $(M_c)_c$ with $0 \le M_c \le 32$, $\sum_c M_c = M$, that satisfies Lemma 2 for all nine blocks. Permuting the symbols of either ternary coordinate, and swapping the two ternary coordinates, are isometries; they act on the grid as a group of order 72, and one representative of each orbit of distributions is kept.

2. **Exact cases.** For each representative, decide by integer programming whether there is a covering code with exactly M_c codewords in each block c: one binary variable per word, one covering constraint per word, and one count constraint per block.

3 Computation

M	distributions	up to symmetry	SCIP 10.0	HiGHS (SciPy 1.16.3)
32	306	10	10 infeasible	10 infeasible
33	1 194	32	32 infeasible	32 infeasible

Table 1: Cases for $K(2, 5, 1)$. Each solver was run on a model built by separate code.

Table 1 summarises the computation. Every case was found infeasible by SCIP 10.0 (through PySCIPOpt) and, with a separately written model, by HiGHS (through SciPy); individual cases took between 1 and about 200 seconds on one core of an Intel Core Ultra 7 255HX.

The following checks were made on the steps before the solvers.

- *Lemma 2 from the definition.* For each block c and each word u, the number of words of c whose ball contains u was computed from Hamming distances alone. It is 6 for $u \in c$, 1 for u in a block of $N(c)$ and 0 otherwise, as in the lemma.

- *Distributions.* A separate, vectorised enumeration of all 76 904 685 ways to place 32 words in nine blocks (95 548 245 for 33), using the coefficients computed above, yields exactly the same 306 (1 194) distributions.

- *Symmetry reduction.* Each of these distributions is equivalent to one of the representatives, and each representative occurs.

- *The exact model.* A 36-word covering code found independently satisfies the model for its own distribution, and the same code with one word removed does not.

- *Known values.* The same pipeline reproduces $K(2, 3, 1) = 12$ and $K(2, 4, 1) = 20$: it reports no code with 11 and 19 words, and finds codes with 12 and 20 words, which an independent brute-force checker confirms.

Remark 3. Both solvers use floating-point arithmetic, so each verdict is evidence rather than an exact certificate; the two solvers come from different teams and agree on every case.

Remark 4. For $M = 34$ there are 4 266 distributions, 93 up to symmetry. SCIP finds 84 of these infeasible; 9 were still open after 300 seconds each. Settling them would give $K(2, 5, 1) \geq 35$.

Code and data. All programs and logs are in the public repository

`https://github.com/Noah-Parsons/covering-codes`

- the search: `src/split_t2.py`;
- the HiGHS recheck: `src/recheck_t2_highs.py`;
- the checks of the reduction: `src/verify_t2_distributions.py`;
- logs: `results/`.

References

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